INCH-POUND

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2 March 1990
SUPERSEDING
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6 December 1979

MILITARY SPECIFICATION

WELDING MACHINE, FRICTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers both continuous drive and inertia type friction welding machines which have the primary function of joining materials of known weldability by using a combination mechanically induced rubbing motion and applied load process with equipment and accessories specified herein.

1.2 Classification. The friction welding machine shall be of the following types. The size required shall be determined by the composition and size of the material to be welded and forge thrust/RPM necessary to successfully friction weld the workpieces. The type and size to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Continuous drive friction welding machine
Type II - Inertial drive friction welding machine

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3438

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SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and Military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS) 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, and publications. The following other Government documents, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B93.114M - Pneumatic Fluid Power - Systems Standard for Industrial Equipment and Machine Tool Applications.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E-18 - Metallic Materials Superficial Hardness, Test for Rockwell Hardness (DOD adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, C/O The National Tool Builders Association, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, NW, Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

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3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operation in accordance with the requirements herein. The machine shall be designed to orient the workpieces concentrically with the axis of rotation between the headstock and tailstock, then to rub the workpieces together and apply the desired forge force for the required time to insure a sound friction weld. The Type I machine shall friction weld by using the continuous drive method of rotating one workpiece, advancing the rotating workpiece interface to contact the stationary workpiece interface, and applying a thrust load. The Type II machine shall friction weld by using the inertial drive method of rotating one workpiece to the required revolutions prior to releasing the stored energy of a disengaged flywheel along with the thrust force being applied to the interface of the two workpieces being joined. The machine shall be designed to operate in ambient temperatures ranging from 41°F to 104°F and relative humidities ranging from 20 percent to 95 percent non-condensing. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Pressure device. A device shall be provided whereby the operator can easily read the applied forge force (pressure) at the weld interface from the normal work position. The device shall be accurate within ± 1% and shall be capable of being recalibrated.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, electrical, and hydraulic operating controls shall be located convenient to the operator's work station.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform with all requirements of 29 CFR 1910. Additional safety and health requirements shall be as specified in detail (see 6.2). The

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friction welder shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial friction welding machine supplied to the commercial market. Additional guarding dependent on workpiece size and configuration shall be provided when specified and fully described by the procuring activity (see 6.2).

3.2.6 Mercury restriction . The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine, and its components, materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. The spindle shall have an automatic lubricating system.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the friction welding machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

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3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a frame and base, headstock, tailstock, drive, ram, hydraulic system, and an electrical system.

3.4.1 Frame and base. The frame shall support all components specified herein. The frame structure shall be either cast iron, cast steel, or a fabricated steel weldment (stress relieved) of the mass, weight, and rigidity to withstand stresses produced when operating the machine at its maximum thrust loads without permanent deformation or failure. Hardened machine ways shall be provided for headstock or tailstock travel. The machine configuration (see tables I and II) of the frame and base shall be specified (see 6.2) as either vertical or horizontal. The frame shall be a floor mounting unit. The base of the frame shall have means to permit leveling and fastening the machine to the floor or foundation. The frame enclosure shall have covered access openings of the size required to facilitate servicing, maintenance, and adjustment of all components contained therein. Eyes or attachment points shall be provided at or near the machine's balance point for lifting the machine. The eyes or lift points shall be capable of supporting five times the weight of the machine, and its location shall not cause permanent distortion when suspended.

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TABLE I. Type I. Continuous drive, friction welding machine, size and capacity.

Size	Capacity				
	Forge thrust (lbs.)	RPM (max.)	Solid diameter 1018 steel (max. in.)	Tubular weld area 1018 steel (max. $in.^2$)	Machine configuration
5	10,000	3,000	1.00	1.07	Horizontal or vertical
6	20,000	3,000	1.13	1.43	Horizontal or vertical
7	30,000	2,500	1.5	2.0	Horizontal or vertical
8	60,000	2,000	1.9	4.2	Horizontal or vertical
9	90,000	1,500	2.39	6.4	Horizontal or vertical
10	120,000	1,300	2.9	8.0	Horizontal or vertical
11	200,000	1,000	3.57	14.1	Horizontal or vertical
12	250,000	1,000	4.0	17.8	Horizontal
13	300,000	1,000	4.6	20.8	Horizontal

NOTE: Machine requirements shall be not less than those stated for the size ordered nor greater than the requirements of the next larger size shown. Machines exceeding these requirements may be considered, but it shall be at the discretion of the procuring activity to determine acceptability of non-complying offers.

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TABLE II - Type II, Inertial drive friction welding machine, size and capacity.

Size	Capacity					
	Forge thrust (lbs.)	RPM (max.)	Flywheel inertia energy (max. lbs.-ft.^{2})	Solid diameter 1018 steel (max. in.)	Tubular weld area (max. in.^{2})	Machine configuration
13	500	45,000/60,000	0.015	0.25	0.07	Horizontal
14	9,000	12,000/24,000	2.25	0.60	0.66	Horizontal
15	13,000	12,000	5	0.90	1.0	Horizontal
16	28,000	8,000	25	1.20	1.7	Horizontal or vertical
17	50,000	8,000	50	1.50	2.6	Horizontal or vertical
18	80,000	8,000	100	2.10	4.6	Horizontal or vertical
19	130,000	6,000	600	2.30	6.5	Horizontal or vertical
20	200,000	4,000	2,500	2.75	10.0	Horizontal or vertical
21	250,000	3,000	5,000	3.125	12.0	Horizontal or vertical
22	350,000	2,000	10,000	3.5	18.0	Horizontal
23	600,000	2,000	25,000	4.5	30.0	Horizontal
24	850,000	1,000	250,000	6.0	42.0	Horizontal
25	1,500,000	1,000	500,000	7.5	75.0	Horizontal
26	4,500,000	500	1,000,000	13.0	225.0	Horizontal

NOTE: Machine requirements shall be not less than those stated for the size ordered nor greater than the requirements of the next larger size shown. Machines exceeding these requirements may be considered, but it shall be at the discretion of the procuring activity to determine acceptability of non-complying offers.

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3.4.2 Headstock. Unless otherwise specified (see 6.2), the headstock shall hold and rotate the workpiece along a centerline between the headstock and tailstock. The headstock shall be constructed with the necessary rigidity and size to absorb the thrust and torsional loads developed by the machine when operating at maximum capacity. The headstock shall contain a spindle which is adaptable for retaining work holding chucks and collets. The spindle opening shall be hardened to a Rockwell C scale range of 53 to 62. The spindle shall deflect not more than 0.002 inch total indicator reading (T.I.R.) under full thrust and highest spindle speeds. The headstock shall be fully supported by anti-friction bearings. The maximum size of the workpieces to be held in the headstock shall be specified (see 6.2).

3.4.3 Tailstock. Unless otherwise specified (see 6.2), the centerline of the tailstock shall be in alignment with the centerline of the headstock. The tailstock shall be sufficiently rigid to maintain this alignment under maximum forces generated on the work by the friction welding machine. The tailstock shall be machined to accept chucks, clamps, or collets for securely holding the stationary workpiece. The tailstock opening or insert shall be hardened to a Rockwell C scale range of 53 to 62. The tailstock opening or insert shall not deflect more than 0.002 inch T.I.R. under full thrust and highest speeds. The tailstock shall travel on hardened and finished surfaces, and provide for the positioning of the tailstock within the machine's range of travel. The maximum size of the workpieces to be held in the tailstock shall be specified (see 6.2).

3.4.4 Drive. Unless otherwise specified (see 6.2), the headstock shall be driven by electrical, mechanical, or hydraulic means. The drive system shall be capable of attaining the specified RPM. When only one type drive is required, it shall be as specified (see 6.2). Revolutions per minute (RPM) shall be repeatable to not more than ± 10 RPM or 1/2 percent of the set value, whichever is greater. The drive shall have a braking system that provides adequate deceleration to assure proper operation of the machine. When a gear set is required for varying the RPM requirements of the machine, the set shall be furnished as standard equipment. Change over of gear sets shall be within the level of routine maintenance procedures. The type II machine shall have a flywheel that has provisions for adding or subtracting weights as required to maintain dynamic balance and to achieve the design revolutions per minute (RPM) performance characteristics.

3.4.5 Ram. Unless otherwise specified (see 6.2), the ram shall be a part of the headstock or tailstock assembly. When only one location of the ram is acceptable, it shall be as specified (see 6.2). The ram shall provide the thrust forces necessary to accomplish the forge bond. The thrust forces shall be variable and controllable. The thrust forces must be repeatable to not more than ± 50 pounds per square inch (psi) or 3 percent of the set value, whichever is greater. Means shall be provided for guiding the travel of the ram and for preventing twisting of the ram under torque loads. The length and type of material to be upset shall be as specified (see 6.2).

3.4.6 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of JIC H-1 and shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Over pressure protection shall be provided in the high pressure line to prevent damage to components.

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A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, provisions for draining, and a cleanout access plate.

3.4.7 Electrical system. The electrical system shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be wired to draw all of its electrical power from a single 230/460-volt, 3 phase, 60-Hertz (Hz) circuit. Unless otherwise specified (see 6.2), the friction welding machine shall be initially wired for operation on 460 volts.

3.4.7.1 Motors. Motors shall meet the requirements of 3.4.7. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.7.2 Programmable controller. The friction welding machine shall be controlled by a programmable controller. The programmable controller shall be designed for use in an industrial environment and implemented in accordance with the original manufacturer's recommendations. The programmable controller shall control all machine functions necessary to friction weld the workpieces automatically by storing program information in a non-volatile memory. Manual operation shall be accomplished by the operator using the control panel. The control panel shall be located convenient to the operator's normal work station. The controls shall include as a minimum, switches, coded energized lights, a spindle speed indicator, a thrust force indicator, a cycle start and stop indicator, and a red, palm type, emergency stop button. The controller shall operate on 120 volts, single phase, 60 Hertz (Hz) input power. The input power shall vary not more than $\pm$ 10 percent.

3.4.7.3 Worklight When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the working area.

3.4.7.4 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the friction welding machine.

3.4.8 Pneumatic system. The pneumatic system shall conform to the requirements of ANSI B93.114M.

3.4.9 Enclosures. The electrical and drive components shall be enclosed. Enclosures shall provide the necessary protection and support for all components operating at maximum capacity. Enclosures shall include means to facilitate ease of maintenance such as access doors and panels. Framing and bases for all enclosures and components shall be adequate to maintain original alignment and to minimize deflection.

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3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the friction welding machine shall meet the requirements of tables I or II for the type and size specified.

3.5.1 Physical envelop limitations. The overall physical envelop of the machine shall be limited, when specified (see 6.2). The physical envelop of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its operating location.

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in 3.4.2 and 3.4.3.

3.7 Performance. The machine shall be capable of joining materials of known weldability with a resultant forged bond equal in cross-sectional strength to the weaker parent material. The machine shall be capable of performing friction welding operations in the workpiece diameters specified in tables I or II for the type and size machine required.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial friction welding machine shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. A corrosion resistant lubrication plate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

3.11.2 Nameplate. A corrosion resistant nameplate shall be securely attached to each friction welding machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below:

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a corrosion resistant hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Type of fluid (SAE number or military or Federal specification number)
Type of filter

3.12 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein and shall be of a quality equal to that of the highest quality products furnished to the commercial market.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The friction welding machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.3. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. An acceptance test shall be performed on the friction welding machine to ensure conformance with this specification. The acceptance test shall be performed at a point designated by the Procuring Activity. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The friction welding machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41^0 to 104^0 Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. The machine shall be operated at no load for not less than one hour. The drive system shall be tested to conform with the requirements of 3.4.4. The operation of the ram shall be tested to conform with the thrust requirements of 3.4.5. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.4.2 and 3.4.3. The alignment tolerance test shall be performed prior to and subsequent to the performance test given in 4.8.4. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

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4.8.4 Performance tests. The machine's performance shall be tested by operating the machine to perform friction welding on the maximum diameter workpieces of AISI 1018 steel to determine conformance with the requirements of 3.7 and tables I and II. Not less than twenty uniform test samples of four inch lengths shall be used in the weld test. Solid diameter test samples shall be friction welded joining the cross-sectional areas into ten weldments. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine. All test weldments shall be concentric end to end within 0.020 inch total indicator reading (T.I.R). The reduction in length in the friction weld area shall be consistent within ± 10%. Not less than five weldments shall be sectioned, polished, etched, and examined at 100 x magnification. The friction weld zone shall be uniform across the weld joint line. The bond shall be uniform and free of unbonded areas or inclusions caused by improper welding. Inclusions present in the parent test material shall be acceptable in the weld area. Not less than five test weldments shall be subjected to destructive tests to determine cross sectional weld strength in the weld area. The cross sectional weld area strength shall be equal to the parent material.

4.8.4.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8.6 Hardness test. The hardness of machine components in 3.4.2 and 3.4.3 shall be tested in accordance with ASTM E 18.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirement for packaging shall be in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The machines covered by this specification are for joining two parts by friction welding. The parts generally are circular in shape and are joined on the ends. The parts may be of the same, similar or different composition, but must be of known compatibility in friction welding.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this solicitation.

b. Type and size required (see 1.2).

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c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Specify system of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5.).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. Specify horizontal or vertical configuration (see 3.4.1).

k. Headstock, if different (see 3.4.2).

l. Specify maximum size of workpieces to be held in headstock (see 3.4.2).

m. Alignment of tailstock with headstock, if different (see 3.4.3).

n. Specify maximum size of workpieces to be held in tailstock (see 3.4.3).

o. Drive system, if different (see 3.4.4).

p. Specify type drive, if required (see 3.4.4).

q. Specify ram's parent assembly, if different (see 3.4.5).

r. Specify location of the ram, if required (see 3.4.5).

s. Specify length and type of material to be upset, if required (see 3.4.5).

t. Electrical system, if different (see 3.4.7).

u. Initial input voltage, if different (see 3.4.7).

v. Motors, if different (see 3.4.7.1).

w. Worklight, if required (see 3.4.7.3).

x. Hour meter, if required (see 3.4.7.4).

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y. Size and capacity, if different (see 3.5).

z. Overall physical envelop of the machine, if required (see 3.5.1).

aa. Additional performance features, if required (see 3.7.1).

bb. Specify and fully describe optional equipment, if required (see 3.9).

cc. Specify applicable repair parts, if required (see 3.10).

dd. Specify information to be provided on lubrication plate, if different (see 3.11.1).

ee. Nameplate, if different (see 3.11.2).

ff. Hydraulic fluid requirements plate, if required (see 3.11.3).

gg. Specify information to be provided on hydraulic fluid requirement plate, if different (see 3.11.3).

hh. Inspection conditions, if different (see 4.3).

ii. First article inspection, if different (see 4.4).

jj. Quality conformance inspection, if different (see 4.5).

kk. Additional performance test, if required (see 4.8.4.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. All technical data furnished should be written in the English language.

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6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a first article sample, a first production item, a sample a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4.8. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification changes. The machine classification changes from the previous issue of this specification reflect the deletion and addition of sizes for type I and II.

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Type I - Continuous Drive

MIL-W-80244A	MIL-W-80244B
Size 1 (deleted)	Size 5 (new size)
Size 2 (deleted)	Size 6 (new size)
Size 3 (deleted)	Size 7 (new size)
Size 4 (deleted)	Size 8 (new size)
	Size 9 (new size)
	Size 10 (new size)
	Size 11 (new size)
	Size 12 (new size)
	Size 13 (new size)

Type II - Inertial Drive

MIL-W-80244A	MIL-W-80244B
Size 1 (deleted)	Size 13 (new size)
Size 2 (deleted)	Size 14 (new size)
Size 3 (deleted)	Size 15 (new size)
Size 4 (deleted)	Size 16 (new size)
Size 5 (deleted)	Size 17 (new size)
Size 6 (deleted)	Size 18 (new size)
Size 7 (deleted)	Size 19 (new size)
Size 8 (deleted)	Size 20 (new size)
Size 9 (deleted)	Size 21 (new size)
Size 10 (deleted)	Size 22 (new size)
Size 11 (deleted)	Size 23 (new size)
Size 12 (deleted)	Size 24 (new size)
	Size 25 (new size)
	Size 26 (new size)

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6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

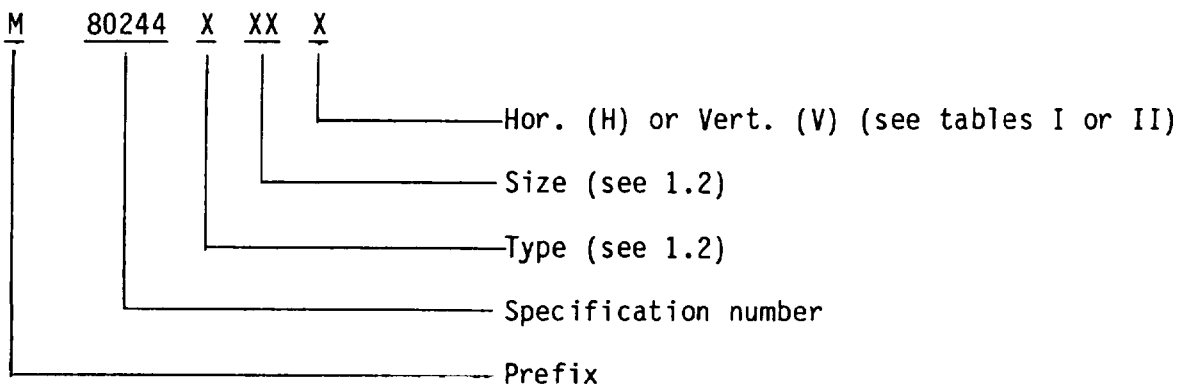
6.9 Safety and health requirements. The 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PIN to be used for the friction welding machine acquired to this specification are created as follows:



6.14 Subject term (keyword) listing.

Continuous drive welding
Forge welding
Friction bonding
Inertia welding
Spin welding

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Custodians:
Army - AL
Air Force - 99

Review activities:
Air Force - 84

User activities:
Army - AV
Navy - MC, AS

Preparing activity:
DLA - IP

Project (3438-0065)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-W-80244B	90 03 02

3. DOCUMENT TITLE

WELDING MACHINE, FRICTION

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198-290

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